Design plan - AI coding

Backend Tech: Python 3.15 + Fast API

Frontend Tech: REACT + Vite + TypeScript

DB: SQLite (Just a DEMO project, select ez DB structure)

AI Coding method: Codex

With the help of AI to generate the backend + frontend code, separate the whole Certificate service to several steps: Each step do test to make sure the developing function is working fine, then jump to next step

Step1: Initialize the project, define the service structure
Step2: DB Design and DATA structure Design - make sure Tenant isolation
Step3: Complete the coding part of Certification list (include web page)
Step4: Complete the Bulk certificate issuance - make sure it can hold broken and retry, cannot reassign certification,
Step5: Complete Certification detail function, support the revoke function for assigned certification
Step6: Complete the CSV export function - Do we need to show valid/revoked in file?
Step7: Complete the AUTO testing for all functions
Step8: Write README and instruction (if need write a script)

Project testing
Using AI to help complete Auto testing, manually testing will focus on the edge case/error retry/Frontend+backend integration test.
Check each modification in code, make sure no dangerous change

Challenge
TimeZone between different tenants: Store timestamps in UTC and use IANA timezone identifiers for tenant-local date handling.
Tenant Isolation, one specific tenant cannot open other tenant data/info

Test edge case
1. Revoke run several times - DB only has one audit log
2. Search List - empty/space/error input
3. CSV export - big table export (such as 100,000 data)
4. Bulk issuance - revoked won't be issued, not enrolled won't be issued, rerun same command will skip all issuance and no extra audit log data
5. Time zone check - store time in DB UTC, the real time is calculated base on UTC time and tenant_timezone